Excitations of Earth's Carbon Cycle

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This manuscript was compiled on May 24, 2019

The history of the carbon cycle is punctuated by enigmatic transient changes in the ocean's store of carbon. Mass extinction is always accompanied by such a disruption, but most disruptions are relatively benign. The less calamitous group exhibits a characteristic rate of change whereas greater surges accompany mass extinctions. To better understand these observations, I formulate and analyze a mathematical model that suggests that disruptions are initiated by perturbation of a permanently stable steady state beyond a threshold. The ensuing excitation exhibits the characteristic surge of real disruptions. In this view, the magnitude and timescale of the disruption are properties of the carbon cycle itself rather than its perturbation. Surges associated with mass extinction, however, require additional inputs from external sources such as massive volcanism. Surges are excited when CO_2 enters the oceans at a flux that exceeds a threshold. The threshold depends on the duration of the injection. For injections lasting a time $t_i \gtrsim 10,000$ yr in the modern carbon cycle, the threshold flux is constant; for smaller t_i, the threshold scales like t_i^{-1}. Consequently the unusually strong but geologically brief duration of modern anthropogenic oceanic CO_2 uptake is roughly equivalent, in terms of its potential to excite a major disruption, to relatively weak but longer-lived perturbations associated with massive volcanism in the geologic past.

Carbon cycle | Mass extinctions | Excitable systems | Dynamical systems | Carbon isotopic events

Introduction

Earth's carbon cycle is a loop between photosynthesis, which converts carbon dioxide (CO_2) to organic carbon, and respiration, which converts organic carbon back to CO_2 (1–4). The cycle has undergone many disruptions throughout Earth history (5–7). These events are expressed in the geologic record as relatively abrupt and large changes in the isotopic composition of sedimentary carbon (8) compared to background fluctuations. Fig. 1 shows two examples in one 800-Kyr time series. Such events are typically attributed to changes in the fluxes and concentrations of carbon (8–10). Thus Fig. 1 could reflect the transient addition of two pulses of isotopically light carbon originating, for example, from the respiration of a previously unreactive reservoir of organic carbon (10, 11). Such disruptions have also been interpreted as geochemical responses to other sources of environmental change, including variations in Earth's orbital parameters (12), dissociation of methane hydrate (13), bolide impacts (14), biogeochemical innovations (15, 16), and changes in chemical weathering (17), organic carbon burial (18), and volcanic emissions (19). These interpretations typically treat the marine carbon cycle as a passive recorder of an externally imposed stress. The response imprinted in the geochemical record is then implicitly assumed to be proportional to the magnitude of the forcing (8, 16, 20).

Given the variety of possible forcing mechanisms, it comes as no surprise that the size and timescale of disruptions vary immensely, by two orders of magnitude during the Phanerozoic (0–542 Ma) (7). Yet the same data show that major fluctuations of the carbon cycle exhibit a characteristic rate of change. Events associated with mass extinction tend to exceed the characteristic rate, whereas others appear bounded by it (7). It seems unlikely that a rich diversity of stressors, expressed as proportionate responses in the geochemical record, would exhibit such uniformity.

Here I formulate and analyze an elementary mathematical model that shows how the characteristic rate can instead emerge *within* the carbon cycle. The model portrays the upper ocean as a chemical reactor open to an incoming flux of dissolved calcium carbonate from rivers and an outgoing flux representing carbonate burial in sediments. Within the reactor, the concentrations of carbonate species respond not only to imbalances in inputs and outputs, but also imbalances in the biological consumption and production of CO_2. This framework reveals a mechanism for autocatalytic amplification of a small but finite perturbation of a stable steady state followed by relaxation back to the same steady state. The process is analogous to the excitation of an action potential (nerve impulse) in a neuron (21). Here, excitations manifest themselves as transient ocean acidification; i.e., a temporary increase in the concentration of carbon dioxide in the upper ocean. This change is distinguished by a characteristic rate.

These results suggest that the magnitude of a carbon cycle's disruption is determined not by the strength of the cycle's perturbation but rather the intrinsic dynamics of the system itself. Once the addition of CO_2 to the oceans passes a threshold, the rate of amplification and the eventual severity of the resulting environmental change should be independent of the detailed history of the perturbation. Moreover, the model indicates that the consequences of fast forcing at human timescales may be similar to the outcome of slow forcing

Significance Statement

The great environmental disruptions of the geologic past remain enigmatic. Each results in a temporary change in the oceans' store of carbon. Although the causes remain controversial, these changes are typically interpreted as a proportionate response to an external input of carbon. This paper suggests instead that the magnitude of many disruptions is determined not by the strength of external stressors but rather the carbon cycle's intrinsic dynamics. Theory and observations indicate that characteristic disruptions are excited by carbon fluxes into the oceans that exceed a threshold. Similar excitations follow influxes that are either intense and brief or weak and long-lived, so long as they exceed the threshold. Mass extinction events are associated with influxes well above the threshold.

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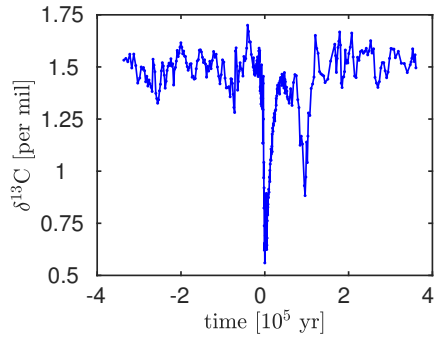


Fig. 1. Fluctuations of the isotopic composition of carbonate carbon (δ^{13}C) during the Eocene period, about 54 Ma (26). Time advances to the right and is given with respect to the minimum of the first abrupt downswing, an event known as Eocene Thermal Maximum 2 (ETM2) or H1. The second event, about 100 Kyr later, is called H2. The time scale is derived from astrochronology (26).

at geologic timescales (7). Within this framework, the greater rates of change associated with mass extinction events are straightforwardly interpreted as the consequence of forcing sustained beyond the threshold, due, for example, to massive volcanism (22–25).

The paper is organized as follows. The first section presents a two-component dynamical system that represents important features of the marine carbonate system. Next, the paper analyzes the stability of the model's steady state. Three dynamical regimes receive focus: the limit cycle that results when the steady state is unstable, the excitations that result when the steady state is stable, and the model's behavior when both the steady state and the limit cycle are stable. Particular attention is given to the case of excitations and their size and timescale. The paper then discusses the implications of these findings for interpreting the geochemical record of past disruptions and predicting the carbon cycle's response to modern anthropogenic perturbations. The conclusion highlights strengths and weaknesses of the paper's principal contributions and points the way towards future progress.

Two-component dynamical system

This section constructs a mathematical model to help explore how instabilities might occur in the marine carbon cycle. Among the many potentially relevant feedback mechanisms (27), I focus on the interactions of shallow-ocean respiration with fluxes of carbon into and out of the shallow ocean. These feedbacks are expressed below as a dynamical system composed of two ordinary differential equations. The objective is to provide a rational context within which one can show how mechanisms operating within the carbon cycle can give rise to general features of its past disruptions. Such understanding follows from the identification of the ways in which the qualitative dynamics of the system—e.g., its stability and bifurcations—can derive from its intrinsic interactions. The theory of dynamical systems makes such connections possible (28–32).

Although these objectives require the identification of relevant mechanisms, they do not necessitate detailed parameterizations. The developments below nevertheless give significant attention to the way in which specific processes operate in the real carbon cycle. I begin by stating essential aspects of carbonate chemistry. The model is then stated in its most general form, after which pertinent mechanisms are expressed mathematically. A final change of variables then leads to a model suitable for classical stability analysis.

The carbonate system. The model expresses the evolution of total alkalinity and the total dissolved inorganic carbon. Total alkalinity is defined by the sum (2, 33)

$$a = [\mathrm{HCO_3^-}] + 2[\mathrm{CO_3^{2-}}] + [\mathrm{B(OH)_4^-}] + [\mathrm{OH^-}] - [\mathrm{H^+}]. \quad [1]$$

Total alkalinity represents the excess of bases over acids. Its dominant contributions come from bicarbonate ions ($\mathrm{HCO_3^-}$) and carbonate ions ($\mathrm{CO_3^{2-}}$); the factor of two attached to the $\mathrm{CO_3^{2-}}$ concentration derives from its double negative charge (33). Total dissolved inorganic carbon (DIC) is symbolized by w, mnemonically recalled as the "whole" of inorganic carbon:

$$w = [\mathrm{CO_2}] + [\mathrm{HCO_3^-}] + [\mathrm{CO_3^{2-}}]. \quad [2]$$

The carbonate system is taken to always be in thermal equilibrium according to the reactions (2, 33)

$$\mathrm{CO_2 + H_2O \rightleftharpoons HCO_3^- + H^+ \rightleftharpoons CO_3^{2-} + 2H^+}. \quad [3]$$

The two conserved quantities (a and w) and the two reactions of Eq. (3) specify the equilibrium of the carbonate system. There are six unknowns: not only a and w, but also the concentrations of $\mathrm{CO_2}$, $\mathrm{CO_3^{2-}}$, $\mathrm{HCO_3^-}$, and $\mathrm{H^+}$. Specification of any two of these quantities therefore provides the other four via equations (1–3) (33).

General formulation. I consider a well-mixed open system in which dissolved calcium carbonate ($\mathrm{CaCO_3}$) is delivered to the shallow ocean by rivers and output from the shallow ocean ultimately into sediments. The assumption of a well-mixed system implies that the model addresses timescales longer than the timescale of global ocean circulation (about 1000 yr (2, 3)). The precise thickness of the shallow layer is unimportant; however it must be small compared to the average depth of the oceans. This assumption allows exchanges of carbon between the shallow ocean and the deep ocean and sediments to be expressed in analogy to the exchange of heat between a small system and a large heat bath. In other words, the concentration of carbon fluctuates within the shallow layer but the mass of carbon below it is effectively constant. Thus the model explicitly considers only the chemical state of the upper ocean.

Fig. 2 illustrates the model. Its mathematical form reads

$$\dot{a} = 2\,[\,j_{\mathrm{in}} - B(a,w)] \quad [4]$$

$$\dot{w} = (1+\nu)j_{\mathrm{in}} - B(a,w) + R(a,w) - (w-w_0)/\tau_w, \quad [5]$$

where dots represent time derivatives. The concentrations a and w are expressed in units of μmol kg^{-1}. The factor of 2 arises because two molar equivalents of alkalinity are associated with each mole of $\mathrm{CaCO_3}$ (33). Expressing the model in terms of the conserved quantities a and w rather than, e.g., the concentrations of $\mathrm{CO_2}$ and $\mathrm{CO_3^{2-}}$, makes it possible to temporarily forestall a discussion of the changing chemical equilibrium.

The flux j_{in} represents the rate at which dissolved $\mathrm{CaCO_3}$ is added to the oceans by rivers in terms of an equivalent change in the whole-ocean concentration of dissolved $\mathrm{CaCO_3}$, expressed in units of μmol kg^{-1} yr^{-1}. The flux $B \geq 0$ denotes

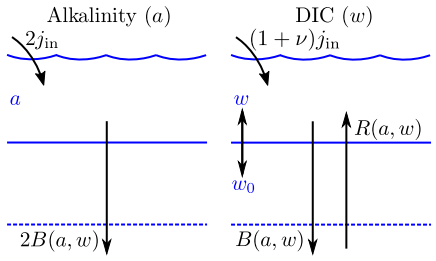


Fig. 2. Schematic diagram of the model (not drawn to scale). The left and right panels represent, respectively, the evolution of total alkalinity, a, and total dissolved inorganic carbon (DIC), w. The wavy line represents the air-sea interface, the upper horizontal bold line divides the shallow ocean from the deep sea, and the dashed line represents the sediment-seawater interface. Concentration fluxes are indicated by unidirectional arrows. The sediment-seawater interface is dashed to indicate that there is no dynamical distinction between the deep sea and sediments.

the rate at which $CaCO_3$ is output from the system, which, in the real ocean, corresponds to burial in sediments. However, the model makes no dynamical distinction between the deep sea and sediments. Thus B represents the rate at which $CaCO_3$ exits the shallow layer without returning. The dimensionless term ν determines the strength of an external source of CO_2, such as volcanic emissions.

The flux $R \geq 0$ represents the extent to which upper-ocean production of CO_2 by respiration exceeds a theoretical baseline. The small change in alkalinity associated with respiration (33) is neglected. The baseline case $R = 0$ is associated with a steady state in which $w = w_0$ and the riverine influx of dissolved $CaCO_3$ is precisely balanced by burial (i.e., $B = j_{\rm in}$). In this case, the model's upper ocean would act merely as a way station for the temporary storage of dissolved carbonate. Positive R, on the other hand, acts to increase w above w_0. The model assumes that this increase is resisted linearly, so that if the baseline $R = 0$ were restored, w would return to w_0 at the timescale τ_w. At long timescales ($\gg 10^2$ yr) in the real ocean, the processes responsible for such dissipation include the riverine influx of dissolved $CaCO_3$ and the dissolution and precipitation of $CaCO_3$ on the seafloor (34, 35). These processes act as the ocean's "homeostat" or "pHstat" at a wide range of timescales (2, 35). Aspects of the homeostat that act simultaneously on DIC and alkalinity respond to the difference between $j_{\rm in}$ and B. However the homeostat must also influence DIC independently by opposing the respiration feedback; otherwise DIC would grow without bound when $R > 0$. Eq. (5) represents this negative feedback by the term proportional to $w - w_0$. The homeostat's dominant characteristic timescale, denoted here by τ_w, is about 10 Kyr (36–38) in the modern ocean.

Equations (4) and (5) describe the general form of a dynamical system in which the riverine influx of alkalinity and DIC produce internal feedbacks that collectively act to determine the extent to which alkalinity and DIC accumulate before exiting the system. The description of the model is completed by the functional specification of the feedbacks B and R, to which we now turn.

The burial and respiration fluxes. The export and burial of $CaCO_3$ in the real ocean depends on the physical environment (e.g., temperature and pressure), the chemical environment, and ecological factors affecting biogenic carbonate precipitation (e.g., phytoplankton community composition). When seawater is under-saturated with carbonate ions, $CaCO_3$ dissolves; when it is supersaturated, $CaCO_3$ can precipitate (33). At low saturations, both physical and biogenic precipitation—and therefore carbonate output—is impeded. The dependence of precipitation on pressure manifests itself as a critical depth beneath which carbonate dissolves. Based in part on earlier work (39), I express these dependencies by

$$B(a,w) = B[c(a,w)] = bj_{\rm in}s(c,c_p), \qquad [6]$$

where the carbonate ion concentration $c(a,w)$ derives from the equilibrium of the carbonate system (33) and

$$s(c,c_p) = \frac{c^\gamma}{c^\gamma + c_p^\gamma} \qquad [7]$$

is the sigmoidal function shown in *SI Appendix*, Fig. S1. The exponent γ parameterizes the sharpness of the transition, c_p is the transitional carbonate ion concentration where $s = 1/2$, and $bj_{\rm in}$ denotes the maximum rate of carbonate burial, The apparent requirement that c be computed from a and w would appear to be a serious complication, but a change of variables discussed below solves this problem. The introduction of the exponent γ may also appear unwelcome; however this paper's conclusions require only that the function s be sigmoidal. Fig. S1 (*SI Appendix*) suggests that $\gamma \simeq 4$ corresponds to the modern ocean.

Respiration is also sensitive to environmental conditions. For example, warmer temperatures should result in increased respiration rates relative to production rates (27, 40–42). Possible consequences include an upward shift of respiration and a positive feedback for shallow-ocean and atmospheric CO_2 levels (27, 41, 43, 44). Alternatively, carbonate production by pelagic calcifiers may decrease at lower pH (45–48). The "ballast hypothesis"(49–51) suggests that the export of organic carbon out of the photic zone is facilitated by its association with biogenic carbonate. A decreased supply of ballast would decrease export of organic carbon and therefore increase respiration rates in the shallow ocean. A positive feedback is again possible (27, 52, 53).

Neither the ballast (54–58) nor the temperature (42, 59–61) feedback is free of ambiguity. However, both are reasonable hypotheses. This paper addresses the case of the ballast feedback, with the expectation that the lessons learned will inform consideration of other possible respiration feedbacks, including those involving temperature.

The essential idea is that lower concentrations of CO_3^{2-} lead to less biogenic carbonate production, and therefore less ballast, less associated organic carbon export, and more upper-ocean respiration (27, 45–53). These observations suggest that the respiration flux R decreases with increasing carbonate ion concentration c. For c below a low concentration of carbonate ions, R should be near its maximum. Conversely, for c above a high concentration, respiration in the shallow layer is assumed to reach its baseline $R = 0$. The baseline case therefore corresponds to the upper-ocean respiration flux that would continue to occur despite elevated production of carbonate ballast.

This reasoning suggests that R decreases with c similar to the way B increases with c (*SI Appendix*, Fig. S1), but with a possibly different amplitude and transitional concentration. I therefore write

$$R(a,w) = R[c(a,w)] = \theta j_{\rm in}\bar{s}(c,c_x), \qquad [8]$$

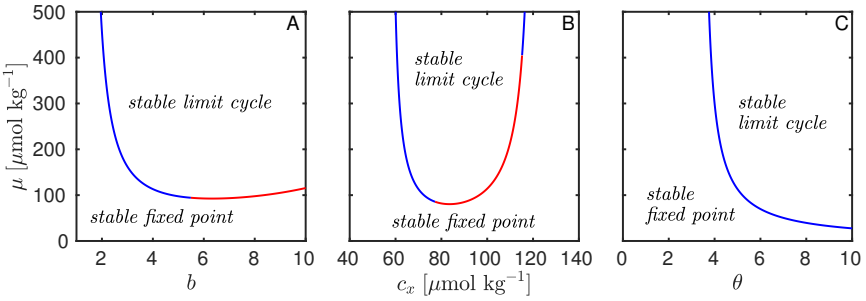


Fig. 3. Stability in the planes of μ and b (A), μ and c_x (B), and μ and θ (C). Values of fixed parameters are given in Table S1. Crossing the red and blue boundaries results in supercritical and subcritical Hopf bifurcations, respectively. The fixed point is stable only where indicated. However the region of stability of the limit cycle extends into the region of the stable fixed point adjacent to the subcritical bifurcation. Fig. 6 maps the bistable region in the plane of c_x and θ.

where $\bar{s} = 1 - s$, θj_{in} is the maximum value of R, and c_x is the "crossover" concentration that marks the halfway point in the transition from high to low R. As for the burial function of Eq. (6), the sharpness of the transition parameterized by γ is a minor detail. Fig. S2 (*SI Appendix*) illustrates how the burial and respiration fluxes of Eq. (6) and Eq. (8) modify Fig. 2.

Transformation from $\dot{a}$ to $\dot{c}$. The *SI Appendix* shows that the rate of change of the CO_3^{2-} concentration can be approximated by

$$\dot{c} = f(c)\,(\dot{a} - \dot{w})\,. \qquad [9]$$

Here the "buffer function"

$$f(c) = f_0 \frac{c^\beta}{c^\beta + c_f^\beta} \qquad [10]$$

approximates the magnitude of the partial derivatives $\partial c/\partial a$ and $\partial c/\partial w$ via specific values of f_0, β, and c_f related to the equilibrium of the carbonate system (*SI Appendix*, Fig. S3). Substitution of Eqs. (6) and (8) into Eqs. (4) and (5) and insertion of the resulting expressions into Eq. (9) then removes the implicit dependence of c on a and w. The new system reads

$$\dot{c}/f(c) = \mu[1 - bs(c, c_p) - \theta\bar{s}(c, c_x) - \nu] + w - w_0 \quad [11]$$

$$\dot{w} = \mu[1 - bs(c, c_p) + \theta\bar{s}(c, c_x) + \nu] - w + w_0, \quad [12]$$

were time is non-dimensionalized by $t \leftarrow t/\tau_w$ and

$$\mu = j_{\text{in}}\tau_w \qquad [13]$$

is a characteristic concentration. The system of equations (11–12) is henceforth called the *carbon-cycle model*. Its parameters are listed in Table S1. Of these, only μ, b, θ, c_x, c_p, and ν are of interest. The remaining five parameters are set to correspond to the equilibrium chemistry or the properties of the modern ocean. Although they are retained in the analysis that follows, they serve only to maintain realism.

Steady state and its stability

Steady states, or fixed points, occur where $\dot{c} = \dot{w} = 0$. Under this condition, the addition of Eq. (11) and Eq. (12) yields $bs(c, c_p) = 1$. Substituting Eq. (7) and solving for c, one then finds the unique steady-state carbonate ion concentration

$$c^* = (b-1)^{-1/\gamma} c_p, \qquad [14]$$

which exists only for $b > 1$. Substitution of c^* for c in Eq. (12) then provides the steady-state DIC concentration

$$w^* = w_0 + \mu\left(\theta + \nu - \frac{\theta c_p^\gamma}{(b-1)c_x^\gamma + c_p^\gamma}\right). \qquad [15]$$

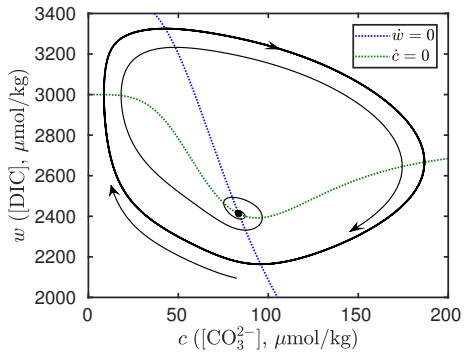


Fig. 4. A stable limit cycle in the phase plane of the CO_3^{2-} concentration c and the DIC concentration w. Values of parameters are given in Table S1. The dotted curves represent the nullclines $\dot{c} = 0$ and $\dot{w} = 0$.

Note that the untransformed model (Eq. (4) and Eq. (5)) yields the same steady state after substitution of Eq. (6) and Eq. (8). The *SI Appendix* shows that parameter values consistent with the modern ocean (Table S1) predict values of c^* and w^* that are also consistent with the modern ocean.

The *SI Appendix* also derives and interprets the results of a linear stability analysis. The requirements for stability are expressed in terms of the trace Tr and determinant Δ of the Jacobian matrix of the system (11–12) evaluated at the fixed point. These calculations yield

$$\text{Tr} = -1 - \frac{\Delta}{2}\left(1 - \frac{\theta b c_p^\gamma c_x^\gamma}{[(b-1)c_x^\gamma + c_p^\gamma]^2}\right) \qquad [16]$$

where

$$\Delta = \frac{2\mu\gamma f_0 (b-1)^{\frac{1}{\gamma}+1} c_p^{\beta-1}}{b(b-1)^{\beta/\gamma} c_f^\beta + b c_p^\beta}. \qquad [17]$$

The fixed point is unstable when Tr > 0. Fig. 3 provides stability boundaries in the planes of μ and b, μ and c_x, and μ and θ. Instability is favored for large $\mu = j_{\text{in}}\tau_w$ because greater rates of riverine input and longer damping times provide for faster growing fluctuations. For some combinations of parameters, however, homeostatic damping always maintains stability.

Limit cycle

The *SI Appendix* shows that the fixed point becomes unstable via a Hopf bifurcation, which leads to periodic nonlinear oscillations called limit cycles. Fig. 4 depicts a stable limit cycle in the phase plane of c and w. Fig. S4 (*SI Appendix*) illustrates the time series $c(t)$ and $w(t)$ on the limit cycle, along with the time evolution of quantities derived from these two quantities.

To understand how the limit cycle operates, consider the evolution of the periodic trajectory from the point where w is at its minimum, i.e., at the lower crossing of the nullcline $\dot{w} = 0$. Here, the ballast feedback begins to increase DIC by the addition of CO_2, which acts to consume CO_3^{2-} ions via the reaction of equation (S8). However the relative rate at which the CO_3^{2-} concentration decreases compared to the rate at which DIC increases becomes progressively slower as the ocean becomes less buffered at lower CO_3^{2-} concentrations. Eventually the CO_3^{2-} concentration reaches a minimum where the loss of CO_3^{2-} via burial and the CO_2 feedback is balanced by the homeostatic response, and it crosses the nullcline $\dot{c} = 0$. However, DIC continues to rise until it is overcome by the homeostatic feedback, reaching a maximum where it crosses the nullcline $\dot{w} = 0$. The homeostatic feedback then acts to to decrease w while increasing c. The CO_3^{2-} concentration reaches its maximum when carbonate burial becomes stronger than the addition of dissolved $CaCO_3$ from rivers and the weakened homeostatic feedback. A rapid upward jump of the burial rate then acts to reduce both the CO_3^{2-} and DIC concentrations. Eventually DIC reaches its minimum and the cycle repeats.

A single period of the limit cycle embodies several attributes of the marine carbonate system that have attracted considerable attention in recent years. The rise of DIC, for example, is initially accompanied by a rise of CO_2 (*SI Appendix*, Fig. S4C) and a reduction of pH (*SI Appendix*, Fig. S4E). This process is called ocean acidification (62). The acid is eventually neutralized by the dissolution of seafloor carbonate and other homeostatic processes, but only slowly, at a modern timescale of about 10 Kyr—part of the long legacy of any rise in CO_2 levels (34). Finally, the surfeit of dissolved $CaCO_3$ must eventually be buried. In the limit cycle pictured here, carbonate burial occurs as a sharp pulse (*SI Appendix*, Fig. S4H) analogous to the "cap-carbonate" deposits that follow global glaciations (63).

Periodic disruptions of the carbon cycle independent of periodic forcing may have occurred in the geologic past (64–67). The following section shows, however, that the potential existence of a limit cycle is more important than its actual existence. Three aspects of the limit cycle will be central to that discussion: its amplitude, its period, and its dynamical origin.

Unlike linear oscillations, the amplitude or size of a limit cycle is independent of the initial condition. As shown in Fig. 4, trajectories originating inside the limit cycle grow outwards, whereas trajectories originating outside move inwards. The resulting amplitude is therefore a property of the system itself rather than its perturbation.

However, the size of the limit cycle does depend on the system's parameters. An appropriate measure of size along the DIC axis is the difference $\Delta w = w_{\max} - w^*$, where $w_{\max}$ is the maximum value of w in the cycle; one similarly has $\Delta c = c_{\max} - c^*$. The *SI Appendix* shows that Δw and Δc generally increase with μ and θ but decrease with b. The limit cycle's size grows roughly linearly with μ because μ sets the rate of all processes other than homeostatic damping. Because rates and size scale similarly with μ, the dimensionless period T of all but the smallest limit cycles should be effectively independent of μ. These observations imply that

$$T \simeq T_0 + 2A/\mu, \qquad [18]$$

where $A = (\Delta w \Delta c)^{1/2}$ is a measure of the amplitude of the limit cycle in units of μmol kg^{-1} and $T_0 \sim 1$ is the dimensionless period of an infinitesimal limit cycle. The factor of 2 reflects an assumption that roughly twice as much time is required to go through a complete cycle compared to the time it takes to change w or c by its typical excursion.

Fig. S5 (*SI Appendix*) shows that equation Eq. (18) is a reasonable approximation. It characterizes the limit cycle well partly because the complicated dependence of A and T on the system's parameters other than μ (*SI Appendix*) is similarly expressed by A and T. The physical interpretation of Eq. (18) is readily exposed by rewriting it in terms of dimensional time. Setting $\tau = T\tau_w/2$, $\tau_0 = T_0\tau_w/2$, and recalling $\mu = j_{\rm in}\tau_w$, one finds

$$A \simeq j_{\rm in}(\tau - \tau_0). \qquad [19]$$

This relationship, which essentially follows from dimensional analysis, shows that, for $\tau \gg \tau_0$, the limit cycle's size is roughly given by the riverine influx $j_{\rm in}$ integrated over the time required for the system to be forced maximally out of its steady state.

The manner in which the limit cycle arises is also of interest. There are two kinds of Hopf bifurcations—subcritical and supercritical (21, 28, 29). The *SI Appendix* performs a standard calculation (21, 28, 29) to determine where each Hopf bifurcation occurs here; the results are color-coded along the curves of Fig. 3. In the carbon-cycle model, the subcritical Hopf bifurcation is adjacent to a bistable region where the stable fixed point and a stable limit cycle appear with an unstable limit cycle. The following section establishes the importance of this region for the interpretation of the model's response to perturbations of the stable fixed point.

Excitability

Now suppose the system is in a stable steady state with no injection of CO_2 ($\nu = 0$). At time $t = 0$ we set $\nu = 0.35$, which corresponds to injecting additional DIC into the oceans at a rate equal to 35% of riverine inputs. From equations (14) and (15), we see that CO_2 injection increases w^* by $\nu\mu$, but c^* is unchanged. The new fixed point retains its stability, because stability is independent of ν (see equations (16) and (17)). However the state of the system must now relax to the new fixed point. Figs. 5A and 5B show that there is a small excursion beyond the fixed point as the trajectory spirals inward to the new steady-state DIC concentration.

Next we repeat the same procedure, but with a slightly larger CO_2 injection rate, $\nu = 0.40$. The initial condition remains the same and the new, stable steady-state DIC concentration is only marginally higher than before. However, Figs. 5C and 5D show that the excursion beyond the fixed point is now roughly an order of magnitude greater than before. Indeed, its size is comparable to the large limit cycle of Fig. 4. The trajectory, however, is not periodic, and the system soon relaxes back to its stable steady state.

Fig. S6 (*SI Appendix*) shows how the size of such excursions varies with ν. The size is defined by the difference between the maximum value of w and w^*_ν, where w^*_ν is the DIC fixed point computed from Eq. (15) for a given value of ν. Fig. S6 (*SI Appendix*) shows that the transition from small to large excursions is sharp yet continuous. Here it occurs near $\nu = 0.391$. Beyond this threshold, the size of the excursion no longer grows appreciably with ν.

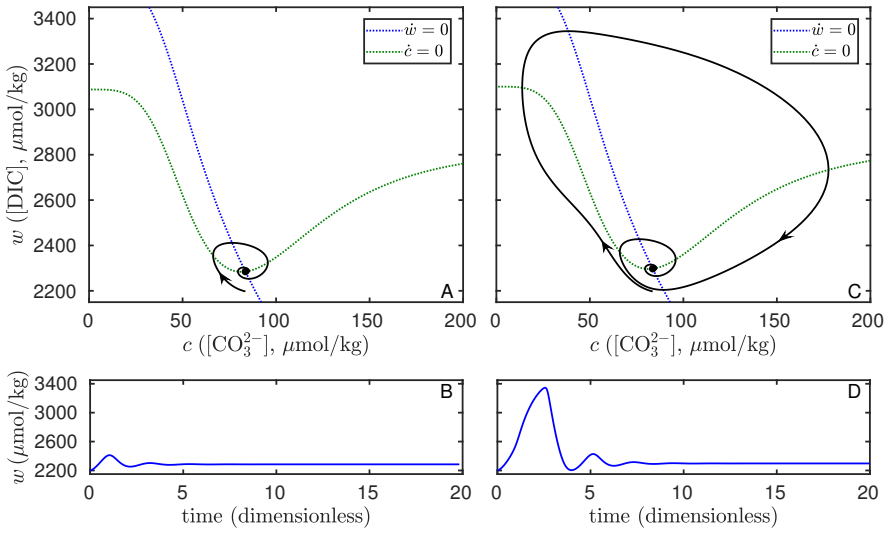


Fig. 5. Evolution following perturbations below and above the excitations threshold. Phase plane representation and the time series $w(t)$ for a sub-threshold perturbation with $\nu = 0.35$ (A and B) and an above-threshold perturbation with $\nu = 0.40$ (C and D). The initial condition and all other parameters are the same in each simulation. The crossover CO_3^{2-} concentration $c_x = 55$ μmol kg^{-1}; other parameters are given in Table S1. The larger value of ν in the above-threshold case increases the DIC fixed point w^* by $0.05\mu = 12.5$ μmol kg^{-1} in (C) compared to (A), which is nearly imperceptible in the plots.

Dynamical systems that exhibit phenomena similar to that of Figs. 5 and S6 (*SI Appendix*) are called *excitable* (21). When a stimulus such as ν exceeds a threshold, excitable systems produce a transient *excitation* significantly larger than the stimulus, as seen here. Excitable systems were first introduced to understand the behavior of action potentials (nerve impulses or spiking) in neurons (68, 69). They have since been studied in the context of numerous other problems, including glacial cycles (32) and the response of peatlands to warming (70).

Excitations occur near Hopf bifurcations (21). Even though the fixed point is stable, a remnant of the stable limit cycle exists in the phase plane. Here, as illustrated in Fig. 5, when ν is above a critical threshold ν_c, the trajectory effectively hops on to a vestige of the limit cycle before settling in to the fixed point. The size and timescale of excitations is therefore comparable to limit cycles. Specifically, their amplitudes are insensitive to the extent to which the threshold parameter exceeds the threshold (*SI Appendix*, Fig. S6), and their duration is comparable to the period of limit cycles. Unlike limit cycles, however, they do not require that a parameter of the system, such as μ, be increased beyond a point of bifurcation. For the carbon-cycle model studied here, this means that excitations can occur in a stable carbon cycle that is merely subjected to an above-threshold injection of CO_2 while its constituent parameters—e.g., riverine fluxes into the system, burial fluxes out of it, and the timescale of homeostatic equilibration—remain the same.

Fig. 6 shows where excitations occur, and at what value of ν_c, in the plane of c_x and θ, along with the stability boundary at which the Hopf bifurcation occurs. Fig. 6 also shows the bistable region adjacent to the stability boundary where the subcritical Hopf bifurcation occurs. One sees that excitations occur in a wide region of parameter space adjacent to the bistable region.

The relation between excitations and bistability is further clarified by Fig. 7, which provides a one-dimensional view along the c_x-axis. The bistable region occurs at values of c_x where the stable fixed point and stable limit cycle coexist with an unstable limit cycle. In this region, values of ν that depress the minimum w of the stable limit cycle below the minimum w of the unstable limit cycle create an immediate jump to the stable limit cycle. At the left boundary of the bistable region, the unstable limit cycle and the stable limit cycle collide via a saddle-node bifurcation of cycles (29). For values of c_x below this point, only the fixed point is stable. Whereas an above-threshold ν creates a jump to the stable limit cycle when c_x is in the bistable region, a similar above-threshold ν creates an excitation at smaller values of c_x. Fig. 8 illustrates how the unstable limit cycle determines the evolution of phase-plane trajectories in the bistable regime. Fig. S7 (*SI Appendix*) shows that excitations near the saddle-node bifurcation can create a finite train of pulses before returning to the stable fixed point.

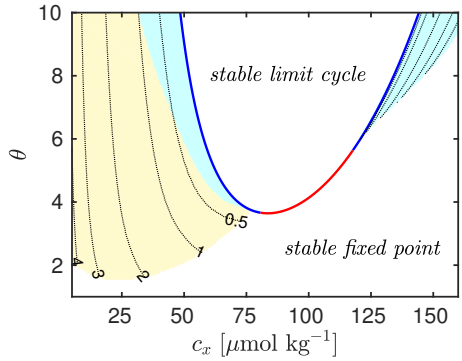


Fig. 6. Contours of ν_c (dotted) in the region of excitations (pale yellow) and jumps to the stable limit cycle in the region of bistability (pale blue) in the plane of c_x and θ. The blue and red portions of the stability boundary correspond to a subcritical and supercritical Hopf bifurcation, respectively. Fixed parameters are specified in Table S1. Excitations or jumps are defined to occur if there exists a ν for which $\mathrm{d}\Delta w/\mathrm{d}\nu > 10^3$, where $\Delta w = w_{\max} - w^*$. The threshold ν_c is the value of ν where that derivative is greatest.

Discussion

If the carbon cycle is indeed excitable, the great events of the geologic past should reveal signs of such dynamics. Among these events, those associated with mass extinction are of particular interest. It is also of interest to ask how the hypothesis of excitability informs our understanding of the risks of human-induced carbon-cycle disruptions in the future. The following discussion provides some first steps in these directions.

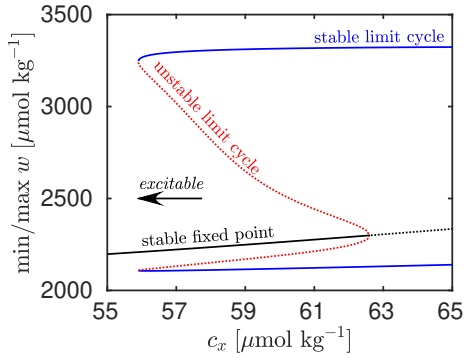


Fig. 7. Bifurcation diagram for the parameter c_x, the crossover CO_3^{2-} concentration for the ballast feedback. The solid and dotted black line respectively represents the stable and unstable fixed point w^*. The solid blue line indicates the maximum and minimum values of w in the stable limit cycle and the dashed red line represents the same extremes of the unstable limit cycle. The subcritical Hopf bifurcation occurs where the radius of the unstable limit cycle goes to zero, at $c_x = 62.61$ μmol kg^{-1}. The saddle-node bifurcation of cycles occurs where the unstable and stable limit cycles collide, at $c_x = 55.89$ μmol kg^{-1}. Excitations occur at smaller values of c_x, to the left of the arrowhead. Fixed parameters are specified in Table S1.

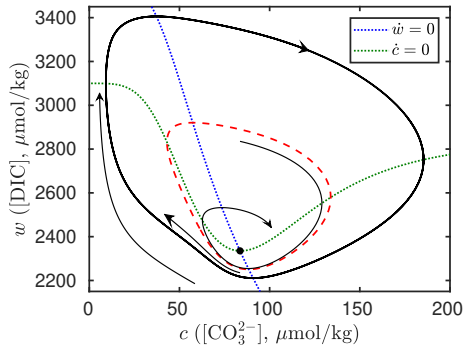


Fig. 8. Phase-space trajectories in the bistable regime. Here $c_x = 58$ μmol kg^{-1} and all other parameters are the same as in Fig. 5C-D. The red dashed limit cycle is unstable; thus trajectories initialized inside it return to the (stable) fixed point. Trajectories initialized outside the unstable limit cycle evolve to the stable limit cycle.

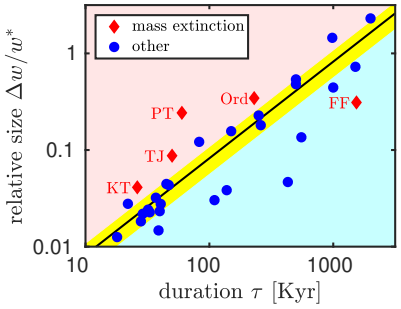


Fig. 9. The relationship between the relative size and duration of 31 disruptions of the global carbon cycle during the last 542 million years (7). The relative size $\Delta w/w^*$ is obtained from changes in the isotopic composition of carbonate carbon that occur in time series similar to that of Fig. 1. The yellow region contains characteristic events; these events satisfy Eq. (20) with $\alpha \sim 0.1$. The events in the pale red region, which include four of the five mass extinction events, grow faster than characteristic events. The light blue region contains minor events with relatively slow growth rates. The labeled events are associated with the end-Cretaceous (KT), end-Triassic (TJ), end-Permian (PT), end-Ordovician (Ord), and Frasnian-Famennian (FF) mass extinctions. Ref. (7) provides descriptions of each event and a discussion of uncertainties.

Characteristic events. Because excitations inherit the properties of limit cycles, Eq. (19) not only relates the amplitude and period of limit cycles via the flux $j_{\rm in}$, but also the amplitude and duration of excitations. As expected, numerical simulations of the carbon-cycle model show that the excitation size $\Delta w \simeq j_{\rm in}\tau$ for large τ, where τ is the time from the onset of an excitation to its peak (*SI Appendix*, Fig. S8). The geochemical record provides data from which the normalized size $\Delta w/w^*$ may be estimated for real events; it also provides estimates of τ (7). If real events are excitations, then the above reasoning suggests that their size and timescale should approximately satisfy

$$\frac{\Delta w}{w^*} = \frac{\alpha j_{\rm in}}{w^*}\tau, \qquad [20]$$

where α is a constant. Because $\alpha j_{\rm in}/w^*$ is a concentration flux divided by a steady-state concentration, I call it the *specific rate* of the excitation.

Fig. 9 plots $\Delta w/w^*$ as a function of τ for 31 global disruptions of the carbon cycle during the last 542 million years (7). Five of the events are associated with major mass extinctions. Variations in τ are likely related to changes in the timescale of homeostatic damping over geologic time (7). The straight line and its uncertainty (the yellow region) predicts the size of the disruption that would occur if it were quantitatively related to the cessation of organic carbon burial and the resulting accumulation of CO_2 (7). Roughly half of the events occur in this region; they each display surges of carbon that grow with approximately the same specific rate. In addition to being well-populated, the yellow region also appears to separate benign events from four of the five mass extinction events. The disruptions within the yellow region therefore appear to share important characteristics in addition to their common specific rate. Henceforth these disruptions are called *characteristic events*.

I hypothesize that characteristic events represent excitations of the marine carbon cycle. If so, then the size of characteristic events is predicted by Eq. (20), just as it is for model excitations. The yellow region of Fig. 9 satisfies Eq. (20) for $\alpha \sim 0.1$ (7). Model excitations scale similarly, but with $\alpha \sim 1$ (*SI Appendix*, Fig. S8). This difference likely reflects mechanisms, such as limitations imposed by organic carbon burial (7), that are not captured by the model. The important point, however, is that for both observed disruptions and model excitations, the amplitude $\Delta w \propto j_{\rm in}\tau$.

The shared scaling of characteristic events and model excitations with the riverine flux $j_{\rm in}$ points to a common foundation. The carbon-cycle model is an open, nonlinear chemical reactor driven out of equilibrium by a flux of reactants into and out of the system. So, too, is the real ocean; elemental concentrations are determined by the balance of inputs and outputs while buffering is controlled by heterogeneous carbonate equilibria (71). More generally, instability, bistability, and oscillations can occur in nonlinear chemical systems forced out of equilibrium (72). In such problems, the size of limit cycles—and therefore the size of any excitations—is a property of the system rather than its perturbation (29). The model (11–12) shows how this works in an idealized carbon cycle. The proportionality of event sizes to the driving flux is a robust property of the model that should be equally applicable in more context settings. The common scaling with $j_{\rm in}$ also provides strong evidence that characteristic events display an intrinsic nonlinear response to perturbation rather than a proportionate response to an external stress. Because different

above-threshold stressors would yield a similar response in an excitable system, the excitation hypothesis also explains why the yellow region of Fig. 9 is relatively well populated. In addition, because the carbon isotopic signals typically return to their initial steady state after each event (e.g., Fig. 1), the steady state apparently remains stable. The combination of these observations is consistent with excitability.

Events lying below the yellow region in Fig. 9 occur at relatively slow specific rates and do not appear to share anything in common other than their slowness. They likely represent a quasistatic (10) response to sub-threshold changes in the parameters of the carbon cycle, such as changes in burial rates. These minor events therefore probably embody the classic proportionate response. The anomalously fast mass extinction events in the upper region should, however, be related to excitability. To understand how, I first discuss the excitation threshold.

The excitation threshold. Practically speaking, the amplitude of an excitation should significantly exceed the amplitude of the perturbation required to exceed the threshold (21). In terms of the carbon cycle model, such a definition immediately provides a conservative upper bound for ν_c by requiring that the increase in w^* due to ν be smaller than the excitation amplitude Δw. Obtaining the former from Eq. (15) and the latter from Eq. (20), one then has $\nu_c \mu < \alpha j_{\text{in}} \tau$ with $\alpha \sim 1$. Similar reasoning applies to the real carbon cycle. Given a damping timescale τ_w, the DIC fixed point should again increase by roughly $\nu_c \mu$ at the threshold. The assumption that characteristic events are excitations then provides the same bound on this increase, but with $\alpha \sim 0.1$. In either case, the excitation timescale τ should be similar to the damping timescale. Recalling $\mu = j_{\text{in}} \tau_w$, one then finds $\nu_c < \alpha$ or

$$\nu_c \lesssim \begin{cases} 1 & \text{model} \quad [21] \\ 0.1 & \text{real world.} \quad [22] \end{cases}$$

Fig. 6 shows that the excitation threshold in the carbon-cycle model is consistent with Eq. (21) near the subcritical Hopf bifurcation.

The combined output of CO_2 from modern subaerial and submarine volcanism produces about 0.06 Pg C yr^{-1} (73), which is about 15% of the modern riverine flux of 0.41 Pg C yr^{-1} (74). If the volcanic flux were doubled, the perturbation expressed by ν would therefore be about 0.15, equivalent to the predicted upper bound for ν_c (Eq. (22)) and therefore enough, in principle, to initiate an excitation.

Mass extinctions. Fig. 9 shows that four mass extinction events exhibit a specific rate significantly greater than that of characteristic events, by a factor of 2–5 (7). These events can be characterized as excitations driven by an injection rate ν significantly greater than ν_c.

The idea follows reasoning similar to that used to bound ν_c. Excitations at threshold exhibit the characteristic size $\Delta w \simeq \alpha j_{\text{in}} \tau$ (Eq. (20)). Suppose that CO_2 injection occurs at rate $(\nu_c + \varepsilon) j_{\text{in}}$, where $\varepsilon > 0$. Equation (15) then shows that Δw will increase by $\varepsilon \mu$ relative to its size at the threshold, resulting in

$$\Delta w \simeq j_{\text{in}} (\alpha \tau + \varepsilon \tau_w). \quad [23]$$

In other words, above-threshold excitations are larger than threshold excitations by a factor of about $1 + \varepsilon \tau_w / \alpha \tau$. Since the excitation timescale τ should not appreciably change, the observed rate would similarly increase. Because $\tau \sim \tau_w$, the main interest lies in ε/α, which is on the order of 10ε for the real carbon cycle. So, for $\nu_c \sim 0.1$ (the upper bound of Eq. (22)), excitation at twice the threshold would increase Δw and its growth rate by roughly a factor of two.

The association of massive flood-basalt volcanism with the end-Permian (23), end-Triassic (22), and end-Cretaceous (24, 25) mass extinctions suggests how this would work. Flood basalts release CO_2 at a rate of about 3.5×10^{-3} Pg C per km^3 of magma (75). Schoene et al. (24) estimate that up to 11 km^3 yr^{-1} of magma erupted from Deccan volcanism over a 10-Kyr period tens of thousands of years prior to the end-Cretaceous extinction. The resulting CO_2 flux is equivalent to about 10% of of the modern riverine flux, yielding $\nu \simeq 0.1$, commensurate with the upper bound for ν_c. Other episodes of massive volcanism (22–25) may exhibit similar pulses (76). Such events may indeed be well above threshold: if ν_c were much smaller than its upper bound, then $\varepsilon = \nu - \nu_c \simeq 0.1$, yielding $\alpha/\varepsilon \simeq 1$. The size and rate of the event would then be roughly double that of characteristic disruptions.

Excitation of the modern carbon cycle. As a result of anthropogenic CO_2 emissions, Earth's oceans currently absorb about 2.3 Pg C yr^{-1} (77), which is about 5.6 times the riverine carbon influx. The resulting excitation parameter, $\nu = 5.6$, is more than fifty times greater than the predicted upper bound for ν_c. That prediction, however, assumes a continuous, constant injection of CO_2. If CO_2 injection occurs instead for a time much less than the damping timescale, its ability to initiate an excitation may be attenuated by the ocean's homeostatic response.

Whether or not excitation ensues will nevertheless still depend on the injection rate. Suppose that CO_2 injection occurs over a time $t_i < \tau_w$. If excitations follow when $\nu > \nu_c$ for $t_i \geq \tau_w$, then excitations should still occur when $t_i < \tau_w$ if the DIC addition $\nu j_{\text{in}} t_i$ is approximately equal to or greater than the total input resulting from injection at the threshold ν_c over one damping time. In other words, there is a t_i-dependent threshold ν_c' that approximately satisfies $\nu_c' t_i = \nu_c \tau_w$ for small t_i. Therefore

$$\nu_c'(t_i) \simeq \nu_c \frac{\tau_w}{t_i}, \qquad t_i < \tau_w. \quad [24]$$

The scaling like t_i^{-1} should increase in accuracy as $t_i \to 0$. Fig. S9 (*SI Appendix*) shows that numerical solutions of the carbon-cycle model compare well to this prediction. The dependence of the short-time threshold on t_i is qualitatively similar to the notion of a critical ramping rate (70).

Given the real-ocean upper bound $\nu_c \lesssim 0.1$ (Eq. (22)), substitution of $\tau_w = 10^4$ yr and $t_i = 10^2$ yr into equation Eq. (24) yields the century-scale upper bound $\nu_c' \lesssim 10$. Mean 21st-century oceanic carbon uptake will likely range from 1.8–4.2 Pg C yr^{-1} (77), which translates to $4.4 \leq \nu \leq 10.2$, similar to the upper bound for ν_c'. With respect to the excitation threshold, the modern perturbation is therefore approximately equivalent to the aforementioned end-Cretaceous pulse; its duration is two orders of magnitude shorter but its flux is two orders of magnitude greater. Fig. 10 illustrates this correspondence graphically.

The short-time critical rate ν_c' may be alternatively expressed as a critical mass m_c. Multiplication of both sides

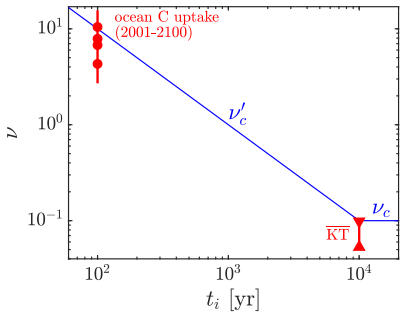


Fig. 10. Equivalence, with respect to the excitation threshold, of the perturbations of the modern and end-Cretaceous carbon cycles, expressed in terms of the dimensionless CO_2 injection rate ν. The straight line labeled ν'_c is the threshold's upper bound predicted by Eq. (24), assuming the modern damping timescale $\tau_w = 10^4$ yr; the segment labeled ν_c (Eq. (22)) provides the upper bound for times $t_i > 10^4$ yr. The circles represent projected 21st-century ocean carbon uptake rates for a range of plausible scenarios (77); the vertical line indicates their uncertainty. The symbols labeled $\overline{\text{KT}}$ provide the median and upper limit of the 68% confidence interval estimated for the period of peak Deccan volcanism tens of thousands of years prior to the end-Cretaceous extinction (24). Because the excitation threshold scales like $1/t_i$, both the modern and end-Cretaceous perturbations potentially lie near the threshold's upper bound despite the two orders of magnitude that separate their rates.

of Eq. (24) by $j_{\text{in}}t_i$ shows that injection with $\nu > \nu'_c$ over a timescale $t_i < \tau_w$ is equivalent to adding a mass of carbon greater than

$$m_c \simeq 12\nu_c j_{\text{in}} \tau_w V, \qquad [25]$$

where $V = 1.35 \times 10^{21}$ kg is the mass of the oceans (2) and the factor of 12 converts mol to gm. Ref. (7) derives an equivalent expression from different assumptions. In conjunction with Eq. (22), Eq. (25) predicts the upper bound $m_c \lesssim 405$ Pg C, which falls within the projected uptake of 290–540 Pg C from the year 1850 to 2100 (77). Thus, either in terms of 21st-century uptake rates or total post-industrial uptake, the modern marine carbon cycle may be at the precipice of excitation.

Finally, there is an important caveat. Since the carbon-cycle model applies only to timescales greater than that of ocean mixing (about 1000 yr (2, 3)), it does not necessarily inform understanding of phenomena at timescales shorter than 1000 yr. However, if the addition of CO_2 to the oceans were not appreciably damped at sub-millenial timescales by processes not discussed here, the reasoning outlined above would remain valid at these short timescales.

Conclusion

Major disruptions of Earth's carbon cycle are typically interpreted as a proportionate response to an environmental perturbation (8, 16, 20). This paper suggests instead that many of these events represent the nonlinear amplification of processes that operate within the carbon cycle. To illustrate how this could work, the paper constructs and studies a simple dynamical-system model of the marine carbon cycle. The results suggest that disruptions can follow when a stable steady state is perturbed beyond a threshold. A likely source of perturbations is an increased flux of CO_2 into the oceans. The autocatalytic amplification of model disruptions exhibits a characteristic increase and rate of growth of the ocean's store of carbon before returning to the original steady state. Such characteristic sizes and rates, independent of the history of external perturbations, are classical features of nonlinear systems (29). The historical record of the real carbon cycle also exhibits similar characteristics (7). These shared properties suggest that the model displays important features of the real carbon cycle.

These findings suggest that a characteristic excitation of upper-ocean CO_2 levels follows the injection of CO_2 into the oceans at an above-threshold rate. If injection continues over a time greater than the timescale τ_w over which the oceans homeostatically adjust to changes in pH, the threshold's upper bound is roughly equivalent to the higher estimates of the rate at which CO_2 degasses from major flood-basalt eruptions associated with mass extinction (24, 76). If injection is instead limited to a shorter duration t_i, the threshold increases by a factor of τ_w/t_i. Thus the relatively slow rate of CO_2 injection commensurate with massive volcanism at geologic timescales turns out to be roughly equivalent, in terms of its potential to reach the threshold, to the much stronger but briefer perturbation of the modern carbon cycle. Mass extinction events appear to be associated with excitations well above threshold.

These conclusions rely in part on the assumption that the dynamics of a two-dimensional dynamical system represent a subset of the complex behavior exhibited by the real carbon cycle. This assumption does not require that the mechanisms encoded by the terms within the dynamical system be strictly correct. However it does demand that the model's qualitative dynamical properties, such as the possible existence of a limit cycle, be realistic. This may not be true. One could argue, for example, that phenomena at sub-millenial timescales, such as ecological reorganization, will act as strong negative feedbacks in local environments, mitigating against global autocatalytic self-organization. The present state of the geochemical record makes such notions difficult to rule out observationally. On the other hand, numerical simulation of more detailed carbon-cycle models (44) should allow one to test whether excitations are expressed in such frameworks. Likewise, careful analysis of individual disruptions or specific periods in the geochemical record should allow one to test whether quantitative signatures of excitable systems exist. The possibility of resonance with the fluctuations of external perturbations (21), such as those induced by orbital variations (12), offers one such route. The work reported here identifies why the carbon cycle may be excitable, how excitability may have been expressed in the past, and why an excitation may occur in the future. In a curious twist of geological and biological evolution, dynamical mechanisms underlying environmental upheaval and mass extinction may be similar to those that make neurons spike.

Acknowledgments

I thank C. Arnscheidt, O. Devauchelle, R. Ferrari, and J. Weitz for helpful discussions, and B. Schoene and B. Keller for providing data. This work was supported by NASA Astrobiology grant NNA13AA90A and NSF grant EAR-1338810.

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Supplementary Information for

Excitations of Earth's Carbon Cycle

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This PDF file includes:

Supplementary text
Figs. S1 to S12
Table S1
References for SI reference citations

Supporting Information Text

Transformation from $\dot{a}$ to $\dot{c}$ via the buffer function $f(c)$. Substitution of Eq. (6) and Eq. (8) into Eq. (4) and Eq. (5) yields

$$\dot{a} = 2j_{\rm in}[1 - bs(c, c_p)] \quad [S1]$$

$$\dot{w} = j_{\rm in}\left[1 - bs(c, c_p) + \theta\bar{s}(c, c_x) + \nu\right] - (w - w_0)/\tau_w. \quad [S2]$$

Figure S2 illustrates this system schematically.

The dependence of c on a and w occurs through complex nonlinear expressions of the carbonate system's thermodynamic equilibrium (1). However, $\dot{c}$ can be obtained from $\dot{a}$ and $\dot{w}$ from

$$\frac{{\rm d}c}{{\rm d}t} = \frac{{\rm d}a}{{\rm d}t}\frac{\partial c}{\partial a} + \frac{{\rm d}w}{{\rm d}t}\frac{\partial c}{\partial w}. \quad [S3]$$

Evaluating the partial derivatives—known in different form as *buffer factors* (2)—also requires knowledge of $c(a, w)$. However a reasonable approximation exists. In general, the partial derivatives have opposite signs (2): increasing alkalinity for constant DIC shifts the equilibrium toward a greater concentration of carbonate ions, whereas increasing DIC for constant alkalinity moves the equilibrium toward a lower carbonate-ion concentration. Figure S3 shows that, as a function of c for a fixed value of w, the magnitudes of the partial derivatives are nearly equal. Moreover they can be approximated by another sigmoidal function. In summary,

$$\frac{\partial c}{\partial a} \simeq -\frac{\partial c}{\partial w} \simeq f_0\frac{c^\beta}{c^\beta + c_f^\beta} \equiv f(c). \quad [S4]$$

Numerical solutions of the carbonate equilibrium show that the positive prefactor f_0, the exponent β, and the transition concentration c_f vary only weakly with w. Here these parameters are taken to be constant and the approximations in Eq. (S4) are treated as equalities. Then

$$\dot{c} = f(c)\left(\dot{a} - \dot{w}\right). \quad [S5]$$

Substitution of Eq. (S1) and Eq. (S2) then provides $\dot{c}$ as a function of c and w. The resulting dynamical system for $\dot{c}$ and $\dot{w}$, identical to Eq. (11) and Eq. (12) of the main text, is

$$\dot{c}/f(c) = \mu[1 - bs(c, c_p) - \theta\bar{s}(c, c_x) - \nu] + w - w_0 \quad [S6]$$

$$\dot{w} = \mu[1 - bs(c, c_p) + \theta\bar{s}(c, c_x) + \nu] - w + w_0, \quad [S7]$$

where $\mu = j_{\rm in}\tau_w$ and time is expressed in units of τ_w.

The buffer function $f(c)$ parameterizes the equilibrium chemistry of carbonate system. While the fit of Figure S3 provides parameters corresponding to the true equilibrium, from a dynamical point of view any sigmoidal shape would yield qualitatively similar results. To understand this from a chemical perspective, add the first and second reactions in Eq. (3) to obtain

$$\mathrm{CO_2} + \mathrm{CO_3^{2-}} + \mathrm{H_2O} \rightleftharpoons 2\mathrm{HCO_3^-}. \quad [S8]$$

If, as in the modern ocean, the concentration of $\mathrm{CO_3^{2-}}$ is much greater than the concentration of $\mathrm{CO_2}$, the addition of one mole of $\mathrm{CO_2}$ (via, e.g., the term ν) consumes nearly as much $\mathrm{CO_3^{2-}}$ to form additional $\mathrm{HCO_3^-}$. In this case, the consumption of $\mathrm{CO_3^{2-}}$ is effectively limited by the concentration of $\mathrm{CO_2}$ and $f(c)$ is near its maximum value. If we imagine instead that the equilibrium $\mathrm{CO_3^{2-}}$ concentration is much lower than that of $\mathrm{CO_2}$ for the same concentration of DIC, the consumption of $\mathrm{CO_3^{2-}}$ is effectively limited by the supply of $\mathrm{CO_3^{2-}}$ and $f(c)$ is much smaller, approaching zero as $c \to 0$. The sigmoidal shape of $f(c)$ arises from these differences at the limits of small and large c, which correspond, respectively, to poorly buffered and well-buffered seawater.

Characterization of the steady state. As one would expect, the steady state $\mathrm{CO_3^{2-}}$ concentration c^* (Eq. (14)) is determined solely by the balance between inputs and outputs, and therefore only the parameters b, c_p, and γ that characterize the burial flux given by Eq. (6). Figure S10A provides contours of c^* as a function of of c_p and b. The average modern deep $\mathrm{CO_3^{2-}}$ concentration is 86 μmol kg^{-1} (3). Taking that as the value for c^* and setting $\gamma = 4$ and $c_p = 110$ μmol kg^{-1} as in Figure S1, one finds $b = 3.7$. The parameter b is the ratio of the maximum rate of carbonate burial to the rate of input of dissolved $\mathrm{CaCO_3}$. It should be similar to the carbonate "overproduction ratio" (4). The overproduction ratio indicates how many times a mole of $\mathrm{CaCO_3}$ cycles between biogenic precipitation and dissolution. Broecker (5) estimates its value to be 4, which is remarkably consistent with the value of 3.7.

The fixed point w^* (Eq. (15)) depends on several additional parameters. Among these, the characteristic concentration $\mu = j_{\rm in}\tau_w$ may be estimated for the modern ocean. The modern riverine dissolved influx $j_{\rm in}$ is about 0.025 μmol kg^{-1} yr^{-1} (6). As discussed in the main text, identification of τ_w with the modern homeostat timescale implies $\tau_w \simeq 10$ Kyr (7–9). Therefore the modern value of μ is about 250 μmol kg^{-1}. The characteristic DIC concentration w_0 represents surface DIC in the absence of a ballast feedback or $\mathrm{CO_2}$ injection; the modern surface DIC concentration of roughly 2000 μmol kg^{-1} (3) is a reasonable choice. The parameters θ and c_x are not straightforwardly measured; they instead represent the combined effects of many biogeochemical processes. Plausible values may be inferred, however, from Figure S10B, a plot of w^* as a function of c_x and θ. The modern deep DIC concentration is about 2280 μmol kg^{-1}, with about 2100 μmol kg^{-1} at a depth of about 100 m. Any combinations of c_x and θ that fall between the contours for 2100 and 2280 μmol kg^{-1} in Figure S10B are therefore consistent with the modern ocean. For example, if $c_x = 55$ μmol kg^{-1}, then θ would lie between 2.5 and 7.1

Linear stability analysis. This section evaluates the stability properties of the fixed point c^*, w^* given by Eq. (14) and Eq. (15) of the main text. It begins by expressing the calculation in its standard form (10–12).

First, shift the fixed point to the origin via the change of variables

$$v = c - c^* \qquad \text{and} \qquad u = w - w^*. \tag{S9}$$

Then rewrite Eqs. (11–12) in terms of v and u to obtain the system

$$\dot{v} = F(v,u) \tag{S10}$$
$$\dot{u} = G(v,u). \tag{S11}$$

Next, form the Jacobian matrix

$$J = \begin{pmatrix} F_v & F_u \\ G_v & G_u \end{pmatrix} \tag{S12}$$

where the partial derivatives

$$F_v = \frac{\partial F}{\partial v}, \quad F_u = \frac{\partial F}{\partial u}, \quad G_v = \frac{\partial G}{\partial v}, \quad G_u = \frac{\partial G}{\partial u} \tag{S13}$$

are evaluated at the origin $(v = 0, u = 0)$. Then compute the trace

$$\tau = \operatorname{tr} J = F_v + G_u \tag{S14}$$

and the determinant

$$\Delta = \det J = F_v G_u - F_u G_v. \tag{S15}$$

The stability of the fixed point depends on the sign of the real part of the eigenvalues λ of the matrix $J - \lambda I$, where I is the identity matrix. The eigenvalues are given by

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}. \tag{S16}$$

It suffices to determine the sign of τ: when $\tau < 0$, the fixed point is stable, and when $\tau > 0$ it is unstable.

Application of the above recipe yields

$$\tau = -1 - \frac{\Delta}{2}\left(1 - \frac{\theta b c_p^\gamma c_x^\gamma}{[(b-1)c_x^\gamma + c_p^\gamma]^2}\right), \tag{S17}$$

where the determinant

$$\Delta = \frac{2\mu\gamma f_0 (b-1)^{\frac{1}{\gamma}+1} c_p^{\beta-1}}{b(b-1)^{\beta/\gamma} c_f^\beta + b c_p^\beta}. \tag{S18}$$

Because $b > 1$ and all other quantities are positive, $\Delta > 0$, and so too is the term proportional to θ in Eq. (S17). One then sees that the condition for instability, $\tau > 0$, can be met, for example, by choosing θ sufficiently large so that the term in parentheses in Eq. (S17) is negative, and choosing μ sufficiently large so that $\tau > 0$. Define μ_c to be the value of μ where $\tau = 0$. At $\mu = \mu_c$, the eigenvalues (Eq. S16) are pure imaginary:

$$\lambda = \pm i \Delta^{1/2}, \tag{S19}$$

where $i^2 = -1$. Increasing $\mu > \mu_c$ then makes real part of λ positive and the fixed point becomes unstable. The same conclusion, via a similar argument, follows if θ is treated as the bifurcation parameter, with θ_c its value at the point of bifurcation.

When the eigenvalues of a two-dimensional dynamical system cross the imaginary axis as they do here, the system undergoes a *Hopf bifurcation*. Determination of whether the Hopf bifurcation is supercritical or subcritical follows the procedure given, for example, on pages 169–170 of Izhikevich (12), page 289 of Strogatz (11), or pages 152–156 of Guckenheimer and Holmes (10). First, following Ref. (12), make a second change of variables

$$v = x \tag{S20}$$
$$F_u u = -F_v x - \omega y, \tag{S21}$$

where the partial derivatives F_u and F_v are again evaluated at $u = 0$ and $v = 0$. This transformation converts the system of Eq. (S10) and Eq. (S11) to

$$\dot{x} = -\omega y + f(x,y) \tag{S22}$$
$$\dot{y} = \omega x + g(x,y) \tag{S23}$$

where

$$f(x,y) = F[v(x), u(x)] + \omega y \tag{S24}$$
$$g(x,y) = -\frac{1}{\omega}\big(F_v F[v(x), u(x)] + F_u G[v(x), u(x)]\big) - \omega x. \tag{S25}$$

Then compute

$$a = \frac{1}{16}(f_{xxx} + f_{xyy} + g_{xxy} + g_{yyy}) + \frac{1}{16\omega}\left[f_{xy}(f_{xx} + f_{yy}) - [g_{xy}(g_{xx} + g_{yy}) - f_{xx}g_{xx} + f_{yy}g_{yy}\right]. \quad [S26]$$

Here the subscripts denote partial derivatives evaluated at $(0, 0)$. For example,

$$f_{xy} = \left.\frac{\partial^2 f}{\partial x \partial y}\right|_{(0,0)}. \quad [S27]$$

When $a < 0$ the bifurcation is supercritical, and when $a > 0$ the bifurcation is subcritical (10–12). The calculation, which is performed using Mathematica, does not yield interpretable expressions. The results are instead expressed graphically by the color-coded stability plots in Figures 3 and 6.

Roughly speaking, instability results when the ballast feedback grows at a faster rate per mole than it is damped. The maximum growth rate is of order $\mu\theta/c_p$ whereas the damping constant is unity (in dimensionless time). The roughly hyperbolic stability boundary in the plane of θ and μ (Figure 3C) therefore corresponds to $\theta\mu \simeq$ constant. Interpretation of the boundary with respect to the crossover concentration c_x is somewhat more subtle. The per-mole growth rate of the ballast feedback is greatest when the CO_3^{2-} concentration $c = c_x$, where the slope of $\bar{s}(c, c_x)$ is steepest. Thus instability is most favored when c_x is near c^*. When c_x is relatively far from c^*, the growth rate is smaller and larger μ is required to outpace the homeostatic feedback. Consequently the boundary in Figure 3B is concave upwards, with its minimum at $c_x = c^*$.

Size of the limit cycle. This section provides approximate results for the dependence of the amplitude of large limit cycles on the parameters of the model. The analysis follows from consideration of the large-c and low-c limits of the nullclines—the curves where $\dot{c}$ or $\dot{w}$ vanish. Setting the right-hand side of Eq. (11) equal to zero, one finds the nullcline where $\dot{c} = 0$:

$$w = w_0 - \mu[1 - bs(c, c_p) - \theta\bar{s}(c, c_x) - \nu] \equiv w_{\rm cn}. \quad [S28]$$

Likewise, Eq. (12) provides the nullcline where $\dot{w} = 0$:

$$w = w_0 + \mu\left[1 - bs(c, c_p) + \theta\bar{s}(c, c_x) + \nu\right] \equiv w_{\rm wn}. \quad [S29]$$

Size along the w-axis. The maximum value of w occurs when the trajectory crosses the low-c end of the w-nullcline. The maximum w can therefore be no greater than the low-c limit of equation Eq. (S29). Let R_w to be the distance along the w-axis from this maximal excursion to the fixed point w^*, normalized by μ:

$$R_w = \frac{1}{\mu}\left(\lim_{c\to 0} w_{\rm wn} - w^*\right) \quad [S30]$$

$$= 1 + \frac{\theta c_p^\gamma}{(b-1)c_x^\gamma + c_p^\gamma}. \quad [S31]$$

The second relation used Eq. (15) and Eq. (S29). This quantity sets the scale of the maximum possible positive perturbation of w from the fixed point w^*. To compare this to numerical simulation, define $w_{\rm max}$ be the maximum observed value of w in a limit cycle. Then

$$\Delta w = w_{\rm max} - w^* \quad [S32]$$

represents the maximum excursion relative to the fixed point for a particular choice of parameters. Figure S11 compares $\Delta w/\mu$, as computed for random choices of μ, θ, b, and c_x, to R_w. We see, as expected, that $\Delta w/\mu \to R_w$ as R_w increases.

Size along the c-axis. The magnitude of the change in w between the small-c limit of the w-nullcline and the large-c limit of the c-nullcline represents a characteristic scale of the change of w as c increases between crossings of the nullclines. Denoting it by R'_w and normalizing by μ, one finds

$$R'_w = \frac{1}{\mu}\left|\lim_{c\to\infty} w_{\rm cn} - \lim_{c\to 0} w_{\rm wn}\right| \quad [S33]$$

$$= |2 + \theta - b|. \quad [S34]$$

To obtain the characteristic change R_c along the c-axis associated with R'_w, write

$$\frac{R'_w}{R_c} = \frac{|v_w|}{v_c}, \quad [S35]$$

where v_w and v_c are characteristic w- and c-components of velocity, respectively, in the phase plane. Although v_w and v_c can vary greatly during one period of a limit cycle, only their characteristic ratio as the trajectory travels between the maximum in w and the maximum in c is required. To a reasonable approximation, this ratio can be viewed as a typical value of the buffer factor $f(c)$, which should be of the order of f_0. Then

$$\frac{R'_w}{R_c} \sim \frac{1}{f_0}, \quad [S36]$$

which yields

$$R_c \sim f_0|2 + \theta - b|. \quad [S37]$$

The carbonate concentration c reaches its maximum $c_{\max}$ when it crosses the c-nullcline $w_{\rm cn}$ for large c. The quantity

$$\Delta c = c_{\max} - c^* \qquad \text{[S38]}$$

is then the maximum excursion of c with respect to the fixed point. Figure S12 compares to Δc to R_c. The nonlinear shape of the function likely indicates a limitation of the validity of Eq. (S36). The constant f_0 will always overestimate the characteristic value of $f(c)$, but as limit cycles become larger and spend more time at high values of c, the approximation should improve. Thus the slope of the function steepens towards unity.

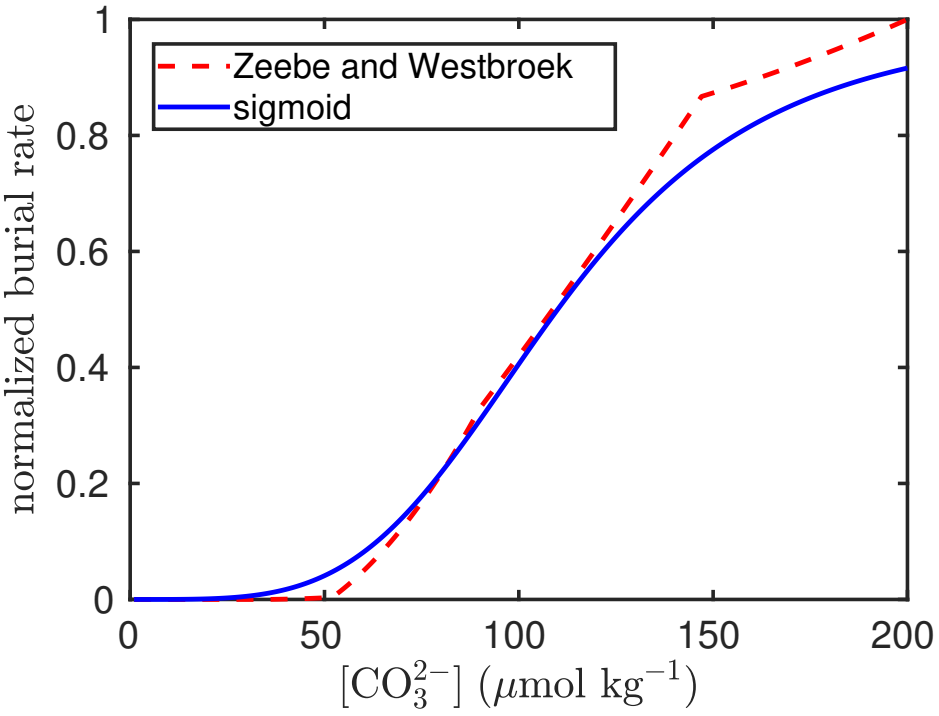


Fig. S1. The piecewise burial function (red dashed lines) of Zeebe and Westbroek (4) compared to the sigmoid function (blue curve) given by Eq. (7), with $\gamma = 4$ and $c_p = 110$ μmol kg^{-1}. Parameters defining the piecewise function are taken from Table 1 of Ref. (4), under the assumption that neritic (shallow water) $CaCO_3$ production is 25% of the total production when $[CO_3^{2-}] = 200$ μmol kg^{-1}. The total output is normalized to unity.

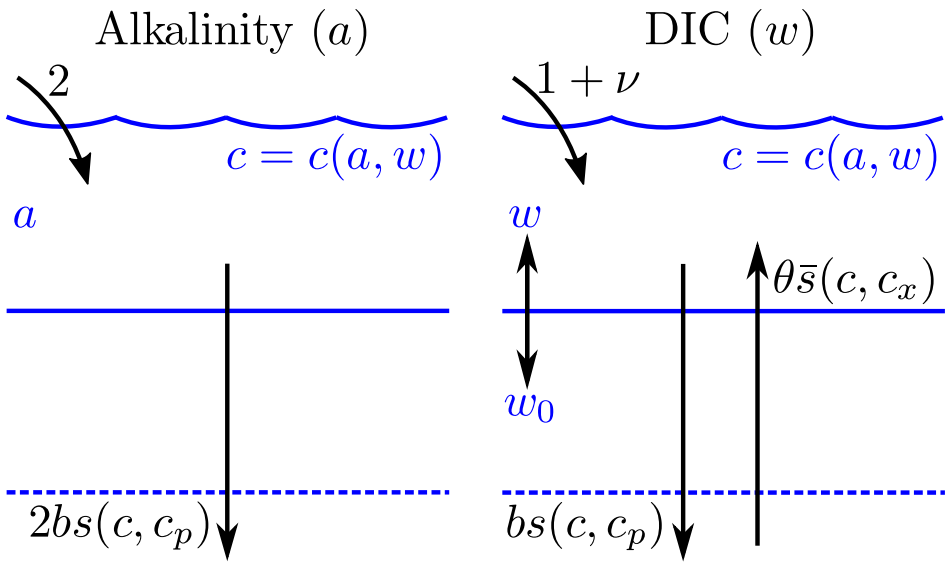


Fig. S2. Schematic diagram of the ballast-feedback model, a special case of the more general model illustrated by Figure 2. Fluxes are indicated in units of j_{in}.

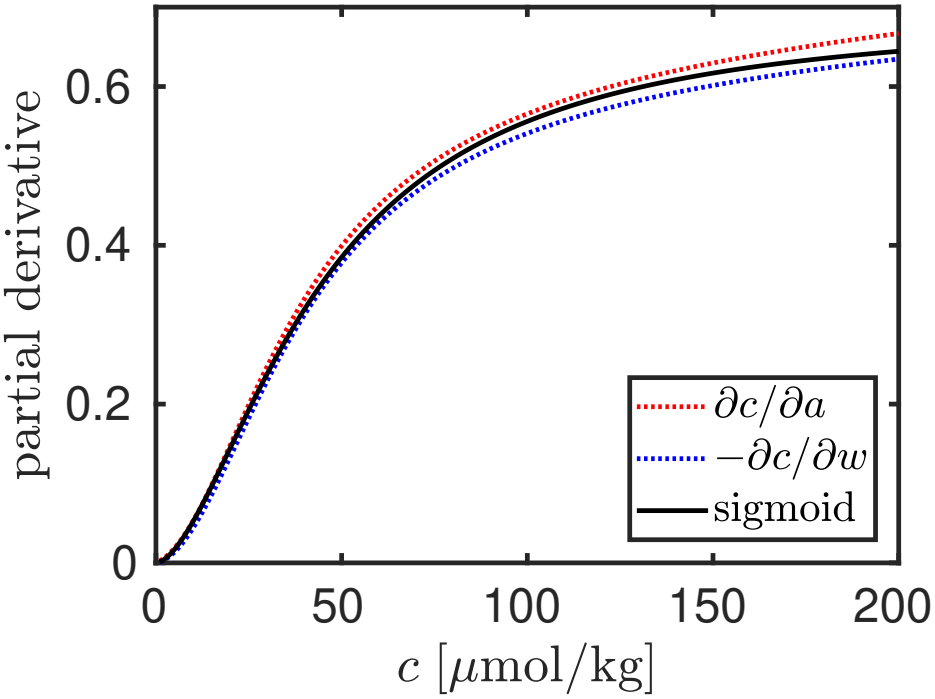


Fig. S3. The partial derivatives $\partial c/\partial a$ and $-\partial c/\partial w$ (dotted lines) as a function of c, computed for $w = 2280$ μmol kg^{-1}, 2.5 °C, 350 bar, and 35% salinity (1). The smooth curve is the approximation given by Eq. (10) of the main text, for $f_0 = 0.694$, $c_f = 43.9$ μmol kg^{-1}, and $\beta = 1.70$.

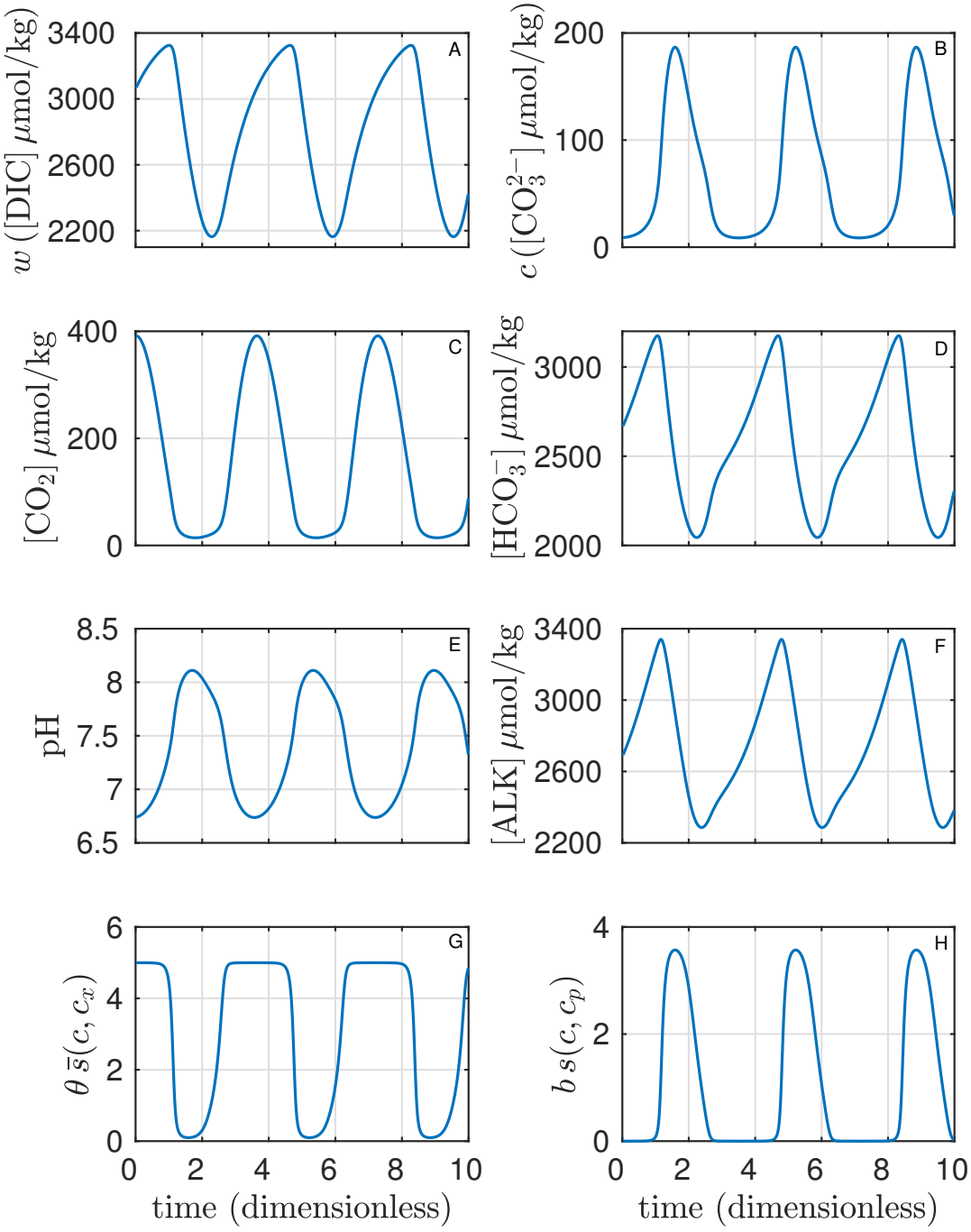


Fig. S4. Time series corresponding to the limit cycle of Figure 4 of the main text. A and B depict the variables w and c, respectively, obtained by integrating Eqs. (11–12) of the main text. The CO_2 concentration (C), the bicarbonate concentration (D), the pH (E), and the total alkalinity (F) are obtained from w and c by solving for the equilibrium of the carbonate system at each instant of time, assuming a temperature of 3.6 °C, a salinity of 35%, and a pressure of 350 bar (1). The dimensionless CO_2 feedback (G) and the dimensionless rate of carbonate burial (H) are computed from $c(t)$. Time is given in units of the homeostat timescale τ_w.

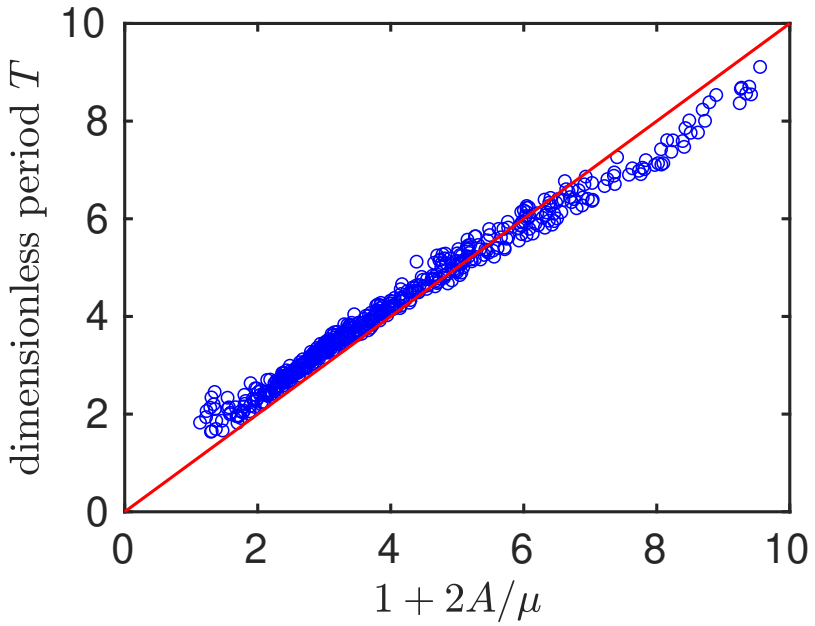


Fig. S5. The dimensionless period T as a function of the rescaled amplitude A of the limit cycle, where $A = (\Delta w \Delta c)^{1/2}$. The circles represent data obtained from numerical simulations using 531 random combinations of μ, θ, b, and c_x that yield a limit cycle with $\Delta w > 10$ μmol kg^{-1}. Here $2 < b < 8$, $2 < \theta < 8$, $150 < \mu < 500$ μmol kg^{-1}, and $65 < c_x < 90$ μmol kg^{-1}. Fixed parameters are given in Table S1. The straight line is the prediction of Eq. (18) with $T_0 = 1$.

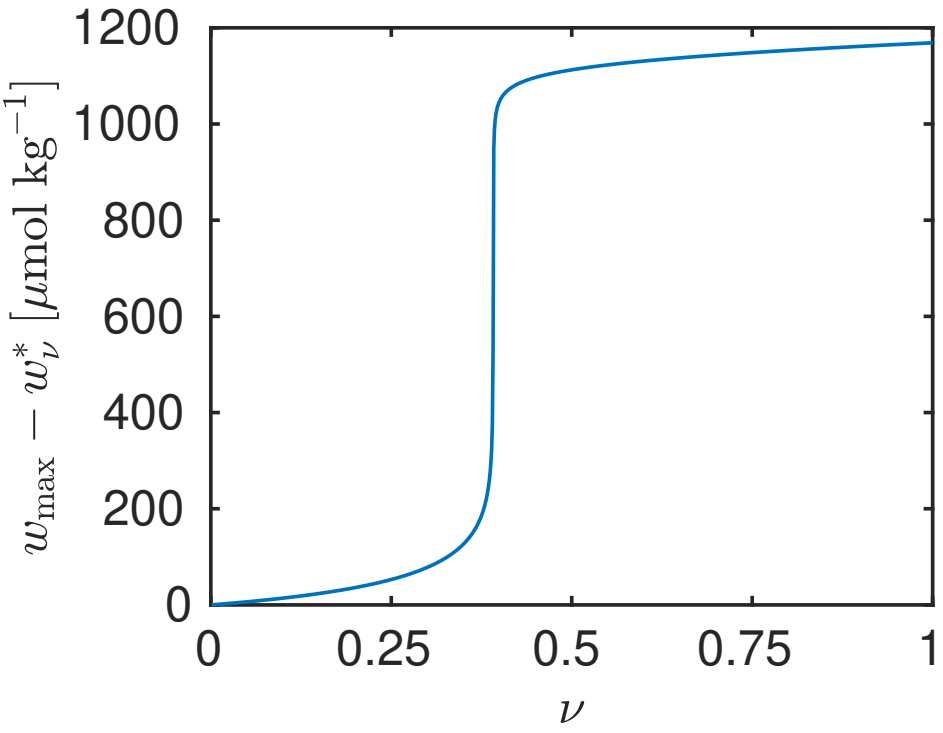


Fig. S6. The maximum excursion of the DIC concentration w with respect to w^*_ν as a function of the dimensionless CO_2 injection rate ν. As in Figure 5, $c_x = 55$ μmol kg^{-1} and other parameters are given in Table S1. The critical injection rate $\nu_c \simeq 0.391$.

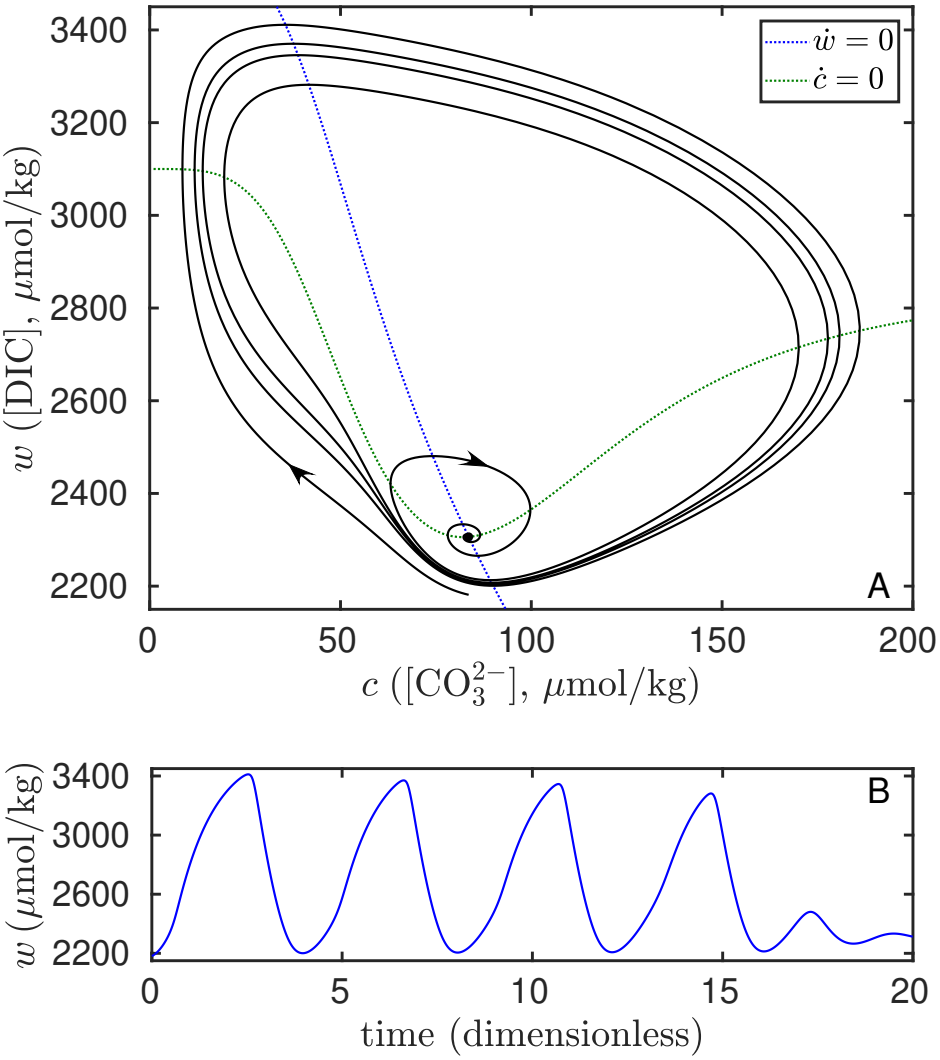


Fig. S7. Excitation near the saddle-node bifurcation of cycles. The phase space trajectory (A) and time series (B) are obtained for an above-threshold excitation with $c_x = 55.73$ μmol kg^{-1}. All other parameters are the same as in Figure 5C-D. Only the fixed point is stable. However, because the system is close to the saddle-node bifurcation of cycles, the excitation results in a series of pulses of similar magnitude before returning to the fixed point.

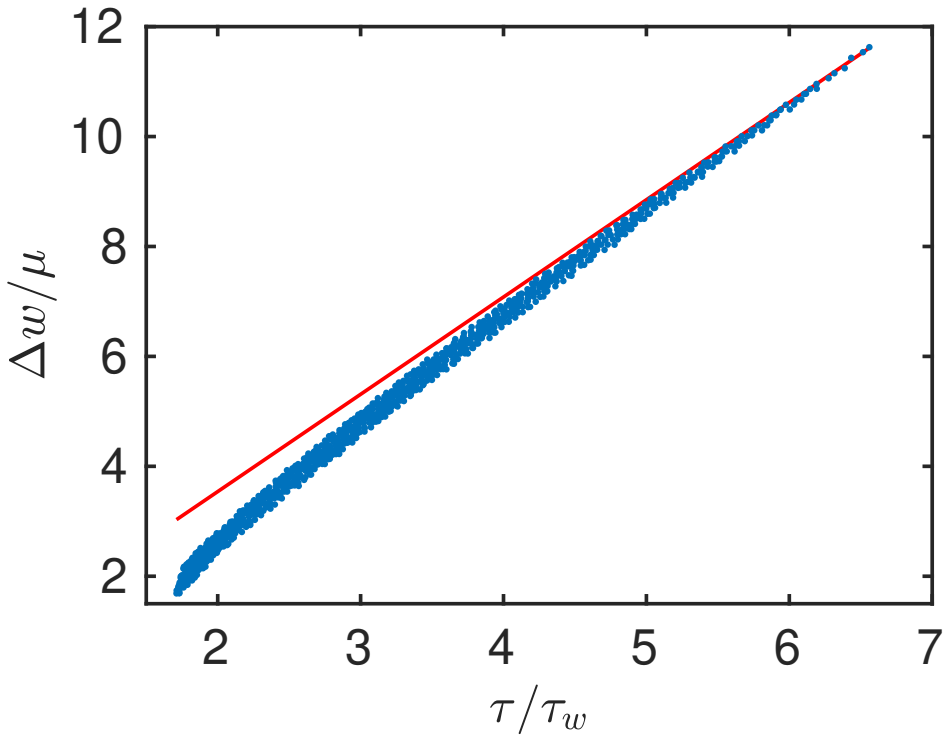


Fig. S8. The dimensionless size $\Delta w/\mu$ (blue dots) of near-threshold excitations in the carbon cycle model as a function of the dimensionless time τ/τ_w from the onset of an excitation to its peak. The red line is a rescaled graph of $\Delta w = \alpha j_{\text{in}} \tau$ with $\alpha = 1.77$. Simulations are performed with varying combinations of c_x and θ that lie in the excitation region of Figure 6 where $\nu_c \leq 1$. Remaining parameters are specified in Table S1.

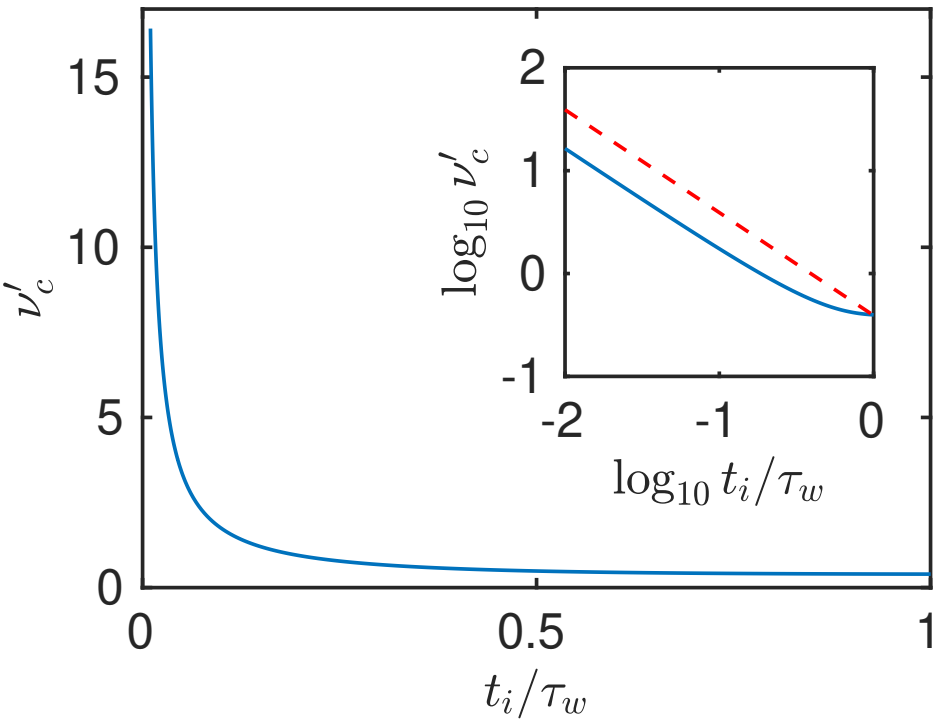


Fig. S9. Dependence of the short-time threshold ν_c' on the injection time t_i in numerical simulations of the carbon-cycle model. Inset: Comparison to the prediction of Eq. (24) on logarithmic axes. Parameters are the same used in Figures 5C and S6.

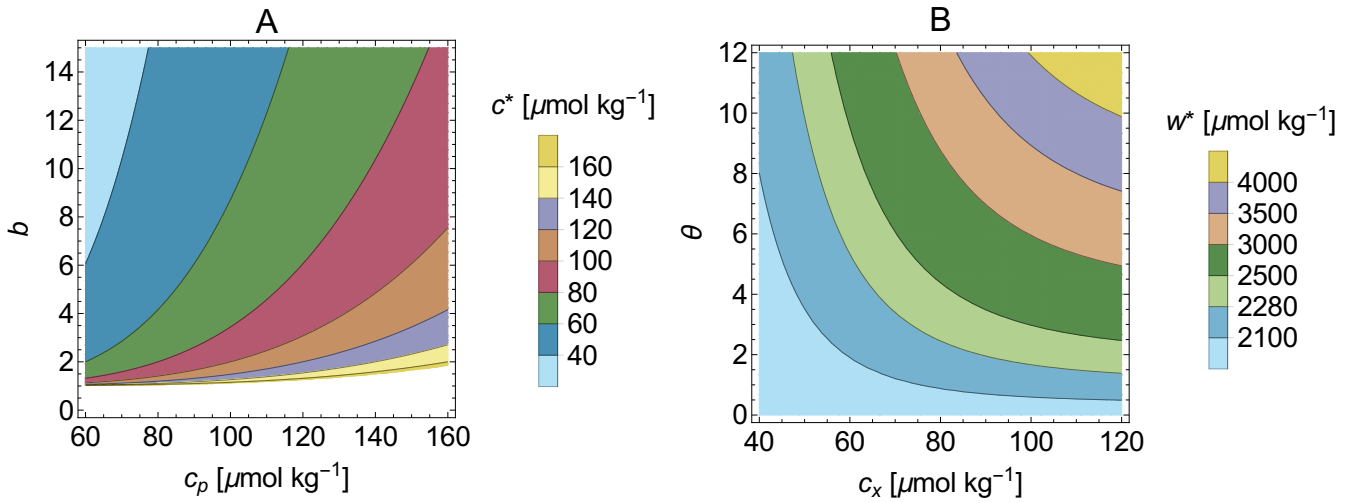


Fig. S10. Steady-state solutions. (A) The fixed point c^* (Eq. (14)) as a function of c_p and b. (B) The fixed point w^* (Eq. (15)) as a function of c_x and θ. Values of unspecified parameters are given in Table S1.

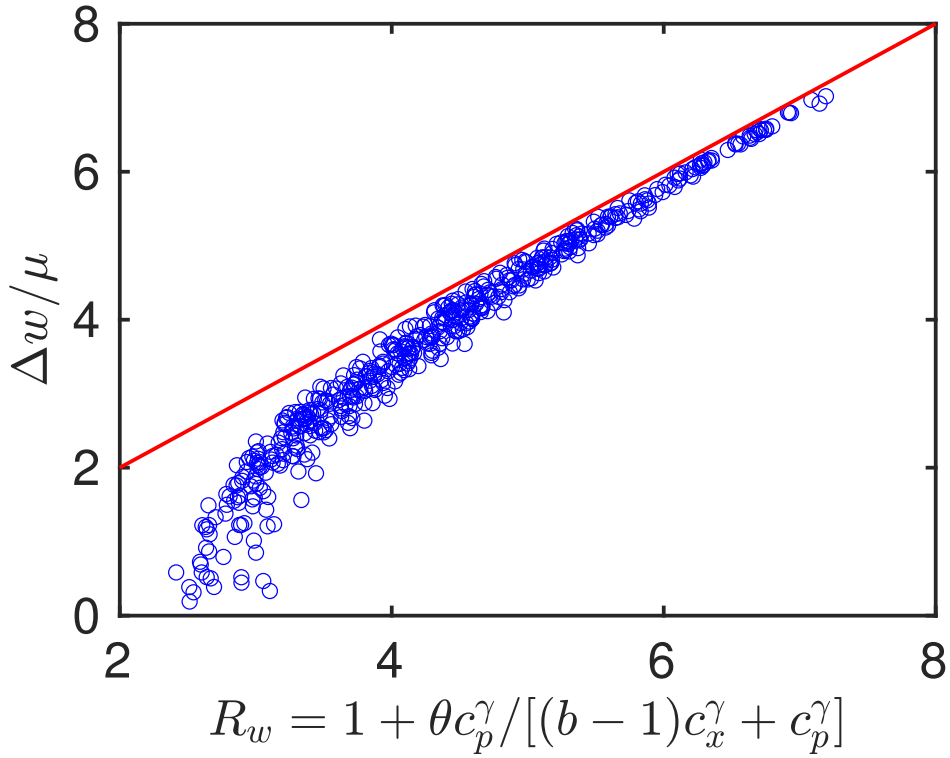


Fig. S11. Comparison of $\Delta w/\mu$, the μ-normalized difference between the maximum level of DIC during a limit cycle and the level of DIC of the unstable fixed point w^*, to the prediction R_w of equation Eq. (S31). The data points represent 531 random combinations of μ, θ, b, and c_x that yield $w_{\max} - w^* > 10\mu$mol kg^{-1}. Here $2 < b < 8$, $2 < \theta < 8$, $150 < \mu < 500\mu$mol kg^{-1}, and $65 < c_x < 90\mu$mol kg^{-1}. The identity line represents equality between the measurement and the prediction of equation Eq. (S31). The theory is asymptotically accurate for large limit cycles. (The Spearman rank correlation coefficient is 0.99.)

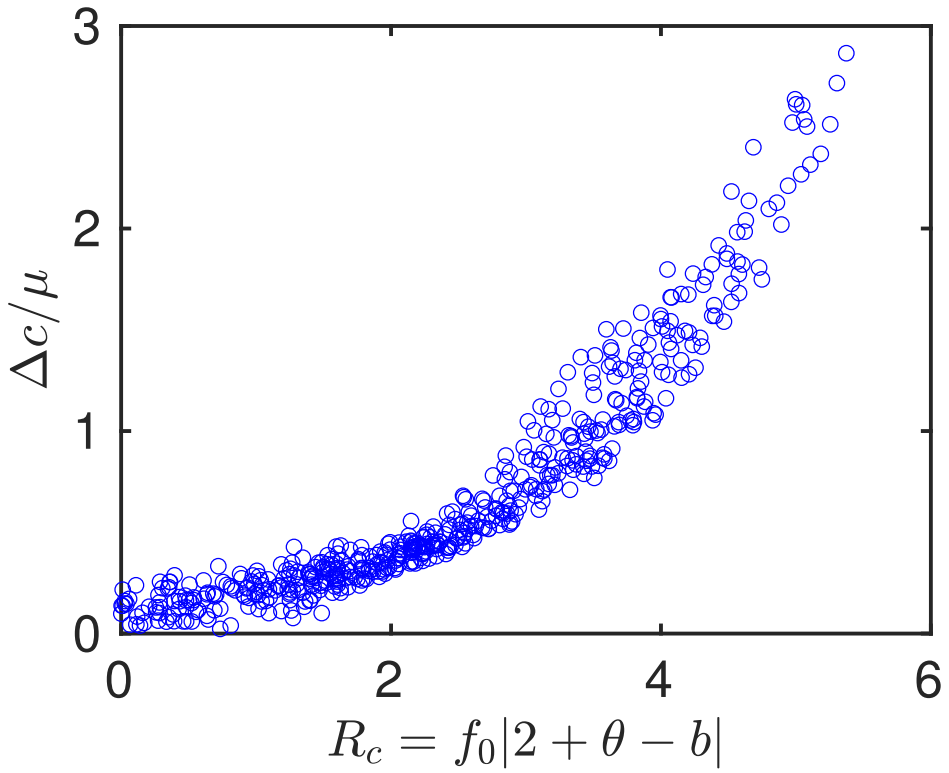


Fig. S12. Comparison of $\Delta c/\mu$, the rescaled difference between the maximum level of CO_3^{2-} ($c_{\max}$) during a limit cycle and the level of CO_3^{2-} of the unstable fixed point c^*, to the scaling of equation Eq. (S37). The data points represent random combinations of the parameters as indicated in Figure S11. The Spearman rank correlation coefficient is 0.97. The curvature reflects changes in the characteristic value of the buffer factor $f(c)$ as the size of the limit cycle increases.

Table S1. Parameters of the carbon-cycle model (Eqs. (11–12)) and their default values.

symbol	value	units	description
μ	250[a]	μmol kg^{-1}	characteristic concentration $j_{\rm in}\tau_w$
b	4	—	maximum $CaCO_3$ burial rate
θ	5	—	maximum respiration feedback rate
c_x	70	μmol kg^{-1}	crossover $[CO_3^{2-}]$ (respiration)
c_p	110[b]	μmol kg^{-1}	crossover $[CO_3^{2-}]$ (burial)
ν	0	—	CO_2 injection rate
w_0	2000	μmol kg^{-1}	reference DIC concentration
γ	4[b]	—	sigmoid sharpness index
f_0	0.694[c]	—	maximum buffer factor
c_f	43.9[c]	μmol kg^{-1}	crossover $[CO_3^{2-}]$ (buffering)
β	1.70[c]	—	sigmoid sharpness index

[a]Obtained from the product of $j_{\rm in} = 0.025$ μmol kg^{-1} yr^{-1} (6) and $\tau_w = 10^4$ yr (7–9, 13).
[b]From Figure S1.
[c]From Figure S3.

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